

scored low here, with the PATA 40 GB being the only sub-50 MBps drive.

The 32 MB tests saw the same four-some taking the top four spots. The Seagate Barracuda 750 GB showed an inexplicable drop in the Average Write part of the 8 MB tests, showing that maybe—just maybe—a drive so large isn't as suitable for writing lots of relatively smaller files.

CPU utilisation isn't a very significant figure today, simply because 4 and 5 percent of CPU resources is a drop of water in a rainstorm for today's multi-gigahertz and dual core processors.

SiSoft Sandra 2007

SiSoft Sandra's latest version gives some nicely detailed scores for individual hard drives. All SiSoft's scores, except the access time, are measured in megabytes per second.

SiSoft Sandra's Drive Index is an average of a drive's scores throughout the gamut of tests it runs, and the highest scorers were once again from the Seagate and Western Digital stables (in that order)—namely the 750 GB 7200.10 Seagate and the WD4000KS. In the 250 GB and below category we saw the Samsung SP2504C lead yet again.

Besides the drive index, SiSoft Sandra also gives figures for Read and Write operations. These are divided into sequential and random figures. Sequential figures measure the drive's ability to read and write data sequentially, that is, to adjacent areas of the disk. This test largely emulates the performance one can expect from a drive while working with large files. Sequential operations are usually done much quicker, and the scores will reflect this.

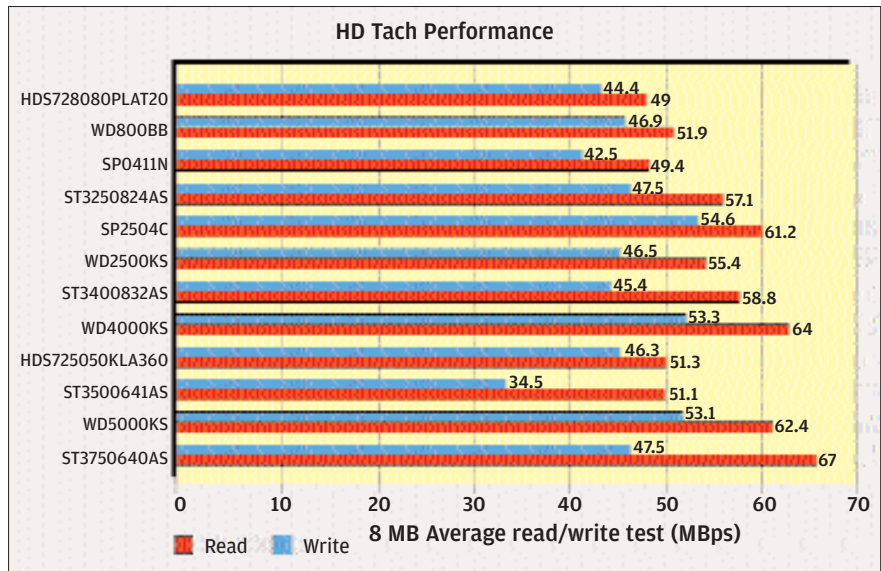
The Random Read and Write scores indicate how well a drive scores when working with scattered data, such as small or discontinuous files. In these tests the 2 heavyweights from Western Digital (the WD4000KS and the WD5000KS), along with the 750 GB Barracuda posted some of the highest read scores we've come across. In the smaller drive group, the Samsung SP2504C once again came out on top.

Among the PATA drives, yet another Samsung drive (for the wrong reasons this time), the 40 GB SP0411N came in last, with the WD1200BB joining it in the under-50 scores for the Drive Index. This is understandable, since the Parallel ATA interface isn't (these days) exactly known for speed. The 500 GB and 250 GB Hitachi drives did quite well.

Real-World Tests

We were more interested in the scores the hard drives would put up in the real-world tests, because these represent actual performance through data transfers both through the bus and through the internal drive buffer.

The file read and file write tests check the I/O performance of a hard drive, as well as its ability to properly utilise the bandwidth that the bus (whether SATA or PATA) provides. A drive that performs better at reading data (whether sequen-



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Hitachi 250 GB
HDT72252DLAT80

tial or random) is more suitable for storing data files such as MP3s and AVIs, that is, non-installable files. This is because these types of files are written only once, but may be read several times. You listen to your music daily, but don't necessarily keep copying it to different locations on your drive partitions, do you?

A drive that excels at writing operations (to find drives suitable for each type of operation, check our benchmarks in the table) is more suitable to use for installing Windows and other program files. This is because the OS and programs are constantly performing write operations in their respective folders. Such drives are also very suitable for creating swap files (which are some of the most write-intensive areas of your entire disk).

The Internal File Transfer involves an intra-drive transfer, where we copy our test files from one partition to another within the same drive. Very suitable to test the performance of a hard drive under stress, simply because data is being read and written simultaneously on the same drive, without the interface between drives playing any significant part. The performance of a hard disk is the sum total of its drive mechanics (including the spindle speed and platter areal density) and I/O buffer. The drive is actually stressed as it has to perform a read and write operation simultaneously, not to mention that the buffer is also being doubly stressed.

The Assorted Internal Read Test was completely dominated by two drives, the Seagate 7200.10 750 GB behemoth, and the slightly smaller (and slower by a hairsbreadth) WD5000KS 500 GB. The only other drive that came anywhere close (read that as a distant third) was the Samsung SP2504C. But this drive was in the smaller capacity group. In all honesty, we were quite disappointed with the Hitachi Deskstar drives, particularly the 7K and T7K series, although no slowcoaches they be; a tad more performance we were hoping to see (if you'll pardon the horrible rhyme!). Among the PATA drives, the WD800BB from Western Digital stood out: it churned out some truly stellar scores, definitely on par with

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Samsung 250 GB
SP2504C